

JONATHAN JONGBIN KIM

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

B.S. in Electrical Engineering

Daejeon, Republic of Korea

Mar. 2025 – Present

GPA: 3.65 / 4.3

Relevant Coursework: Introduction to Reinforcement Learning, Introduction to Generative Models, Machine Learning Basics and Practices, Probability and Introductory Random Processes, Linear Algebra

RESEARCH INTERESTS

Reinforcement Learning, World Models, Generative Models, LLM Post-training

SELECTED PROJECTS

3D Gaussian Splatting — Scene Reconstruction [GitHub]

2026

PyTorch renderer implementation and gsplat-based 3D reconstruction

- Implemented a pure-PyTorch 3DGS renderer with 3D covariance, perspective projection, depth sorting, and differentiable alpha compositing.
- Built a COLMAP-to-gsplat training pipeline with SfM initialization, L1/SSIM losses, progressive spherical harmonics, and densification hooks.
- Added ablation controls for densification, parameter freezing, and scale/opacity regularization, with TensorBoard logging and PSNR/L1 validation.
- Trained on 135 registered views for 30,000 steps on a Tesla T4, achieving validation PSNR of 27.354 dB and L1 error of 0.03535 with 505,695 Gaussians.

StarGAN v2 — Multi-Domain Image Translation [GitHub]

2025

Paper-based implementation and empirical evaluation

- Implemented a PyTorch framework for 256×256 multi-domain image translation on AFHQ, including a generator, mapping network, style encoder, and discriminator.
- Combined AdaIN-based style conditioning with adversarial, style reconstruction, diversity-sensitive, and cycle consistency losses in the training pipeline.
- Trained and evaluated the model on a MacBook Pro with an Apple M4 Pro and 24 GB RAM using the PyTorch MPS backend.
- Evaluated a 20,000-step checkpoint on 3,000 generated images, achieving an overall FID of 44.11 and LPIPS diversity of 0.4270; analyzed domain-specific generation quality and visual changes across training checkpoints.

SELECTED PUBLICATIONS

[1] **J. Kim**, et al., “Analyzing The Impact and Evaluation of Flexitime on Subway Congestion Using SARIMA Model”, *Proceedings of the Korea Computer Congress (KCC)*, 2024.

[2] **J. Kim**, et al., “Comparison of Deep Learning Architecture Performance and Suggestion of Baseline Model for Diagnosing Cassava Leaf Disease”, *Proceedings of the Korea Software Congress (KSC)*, 2024.

HONORS & AWARDS

Excellence Award, Junior Division

2024

Undergraduate/Junior Paper Competition, Korea Software Congress (KSC)

SKILLS

Programming: Python, C++, Bash

Machine Learning: PyTorch, NumPy

Tools: Git, Linux, CUDA, L^AT_EX